

Close

B S CHENTHIL HARI
Decision Tree - Age and Income predicting House Purchase

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```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier, export_text, plot_tree
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
```

▼ Create Synthetic Dataset

The rule: people over 30 are more likely to buy a house (70% of them do). Income provides the second split.

```
import numpy as np
np.random.seed(42)

n = 200
age = np.random.randint(20, 60, n)
income = np.random.randint(20000, 120000, n)

# 70% of age > 30 buy; 30% of age <= 30 buy
buys = []
for a, i in zip(age, income):
    if a > 30:
        buys.append(1 if np.random.random() < 0.7 else 0)
    else:
        buys.append(1 if np.random.random() < 0.3 else 0)

data = pd.DataFrame({'age': age, 'income': income, 'buys': buys})
print(data.head(10))
print("\nClass distribution:")
print(data['buys'].value_counts())
```

```
   age  income  buys
0   58   62107     0
1   48  106202     1
2   34   71663     1
3   27   35708     0
4   40  111906     1
5   58   69811     1
6   38   22811     0
7   42   76250     0
8   30   92082     1
9   30   54754     0
```

```
Class distribution:
buys
1    113
0     87
Name: count, dtype: int64
```

▼ Train Decision Tree

```
x = data[['age', 'income']]
y = data['buys']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=5)

model = DecisionTreeClassifier(criterion='entropy', random_state=5, max_depth=4)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

▼ Print Decision Tree Rules

```
print(export_text(model, feature_names=['age', 'income']))
```

```
|--- age <= 30.50
| |--- age <= 25.50
| | |--- age <= 22.50
| | | |--- income <= 34850.00
| | | |--- class: 1
```

```

| | | | |--- income > 34850.00
| | | | |--- class: 0
| | | |--- age > 22.50
| | | |--- class: 0
| | |--- age > 25.50
| | |--- age <= 26.50
| | |--- income <= 80566.50
| | |--- class: 1
| | |--- income > 80566.50
| | |--- class: 0
| | |--- age > 26.50
| | |--- income <= 91176.50
| | |--- class: 0
| | |--- income > 91176.50
| | |--- class: 1
|--- age > 30.50
|--- income <= 95557.50
|--- income <= 81350.50
|--- income <= 76748.00
|--- class: 1
|--- income > 76748.00
|--- class: 1
|--- income > 81350.50
|--- income <= 86397.50
|--- class: 0
|--- income > 86397.50
|--- class: 0
|--- income > 95557.50
|--- age <= 41.50
|--- class: 1
|--- age > 41.50
|--- income <= 111247.50
|--- class: 1
|--- income > 111247.50
|--- class: 1

```

Visualize Decision Tree

```

plt.figure(figsize=(12, 8))
plot_tree(model, feature_names=['age', 'income'], class_names=['No Buy', 'Buys'],
          filled=True, rounded=True, fontsize=9)
plt.title('Decision Tree: Age & Income -> House Purchase')
plt.tight_layout()
plt.show()

```

Decision Tree: Age & Income -> House Purchase

